

THINKING

Theories · Concept Formation · Thinking Styles · Mental Set

Higher Mental Processes — Academic Psychology

✓ Theories of Thinking

✓ Concept Formation

✓ Thinking Styles

✓ Set in Thinking

Thinking is a higher mental process through which individuals manipulate information, ideas, symbols, and experiences to understand situations, solve problems, make decisions, and generate new ideas. It involves the active processing of information beyond immediate sensory input and plays an essential role in human adaptation and intelligent behavior.

1. Theories of Thinking

Psychologists have proposed different theories to explain how thinking occurs and how people solve problems.

A Association Theory

Associationists believed that thinking consists of a chain of ideas connected through learned associations.

Basic Principles

Ideas become linked through experience according to certain laws:

Law	Description	Example
Law of Similarity	Similar ideas tend to evoke one another.	Seeing a tiger may remind someone of a lion.
Law of Contrast	Opposite ideas become associated.	Thinking about day may lead to thoughts of night.
Law of Contiguity	Events occurring together in time or space become linked.	The smell of popcorn may remind a person of watching movies.

Strengths

- Explains simple learning processes.
- Accounts for habitual thought patterns.

Limitations

- Does not adequately explain creative thinking or reasoning.
- Cannot explain sudden insights.

B Gestalt Theory of Thinking

Gestalt psychologists argued that thinking involves perceiving relationships and organizing information into meaningful wholes.

Main Assumptions

- Thinking is not merely combining separate ideas.
- Individuals perceive situations as organized patterns.

- Problem-solving occurs through understanding relationships among elements.

Insight Learning

Insight is the sudden realization of a solution after understanding the structure of a problem.

Characteristics of Insight	Example
• Sudden appearance of the solution. • Based on comprehension rather than trial-and-error. • Once discovered, the solution is easily remembered.	A student struggling with a mathematical problem suddenly recognizes a pattern and immediately arrives at the correct answer.

Contributions: Gestalt theory emphasized organization of perception, productive thinking, and creativity.

Contributions • Organization of perception. • Productive thinking. • Creativity.	Limitations • Difficult to measure insight scientifically. • Does not explain routine thinking well.
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C Information Processing Theory

This modern approach compares human thinking to the functioning of a computer.

Assumptions: Thinking involves receiving information, encoding information, storing information, retrieving information, manipulating information, and producing responses.

Component	Description
Input	Information enters through sensory systems.
Processing	Mental operations analyze and transform information.
Storage	Knowledge is maintained in memory.
Output	Decisions, actions, or verbal responses are produced.

Advantages • Explains complex cognitive activities. • Supported by experimental research. • Useful in understanding decision-making.	Limitations • May underestimate emotions and social influences on thinking.
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2. Concept Formation

Concept formation refers to the process by which individuals categorize objects, events, or ideas based on shared characteristics.

Concepts help people organize experiences and reduce cognitive complexity.

Examples: Dog, Fruit, Democracy, Intelligence.

Types of Concepts

Type	Description	Examples
A. Natural Concepts	Developed through everyday experiences.	Birds, Trees, Chairs
B. Artificial Concepts	Defined by precise rules.	Triangle, Prime number, Chemical elements

Process of Concept Formation

1	Observation	Individuals encounter various objects or events.
2	Comparison	Common and distinctive features are examined.
3	Abstraction	Relevant characteristics are identified.
4	Generalization	Objects sharing similar attributes are grouped together.
5	Testing	The concept is applied to new examples.

Factors Affecting Concept Formation and Importance

Factors Affecting Concept Formation	Importance of Concept Formation
• Intelligence • Language development • Previous experiences • Motivation • Education • Cultural background	Concept formation enables individuals to: • Simplify information • Improve communication • Enhance reasoning abilities • Facilitate problem-solving • Support learning

3. Thinking Styles

Thinking styles refer to preferred ways individuals process information, approach problems, and make decisions.

Different people may possess different dominant styles.

Style	Description	Characteristics	Example
A. Concrete Thinking	Focuses on observable facts and direct experiences.	Literal interpretation; Preference for practical information; Difficulty understanding abstract ideas.	<i>Following cooking instructions exactly without modification.</i>
B. Abstract Thinking	Involves dealing with ideas, symbols, and hypothetical situations.	Understands metaphors; Uses imagination; Engages in theoretical reasoning.	<i>Discussing justice, freedom, or morality.</i>
C. Convergent Thinking	Produces a single correct answer.	Logical; Structured; Analytical.	<i>Solving mathematical problems.</i>
D. Divergent Thinking	Generates multiple possible solutions.	Creative; Flexible; Original.	<i>Brainstorming uses for a paper clip.</i>
E. Reflective Thinking	Involves careful evaluation before making decisions.	Systematic analysis; Consideration of alternatives; Deliberate judgment.	<i>Comparing several career options before selecting one.</i>
F. Impulsive Thinking	Characterized by rapid responses with limited consideration.	Quick decisions; Less evaluation; Greater likelihood of errors.	<i>Purchasing an expensive item without comparing prices.</i>

4. Set in Thinking (Mental Set)

A mental set is a tendency to approach problems using methods or solutions that have been successful in the past.

It represents a habitual way of thinking that can facilitate or hinder problem-solving.

Characteristics of Mental Set

- Based on previous experiences.
- Produces expectations about solutions.
- Influences perception and reasoning.
- May reduce flexibility.

Advantages

- Saves time and effort.
- Promotes efficiency in familiar situations.
- Enables rapid decision-making.

Disadvantages

- May prevent discovering better solutions.
- Encourages rigid thinking.
- Can interfere with creativity.

Example: A student who always solves algebra problems using one formula may continue applying the same method even when a simpler approach is available. This persistence illustrates a mental set.

Conclusion

Thinking is a complex cognitive process that enables humans to reason, solve problems, make decisions, and adapt to changing environments. Theories of thinking explain the mechanisms underlying thought processes, concept formation organizes knowledge into meaningful categories, thinking styles reflect individual preferences in processing information, and mental sets demonstrate how past experiences can both facilitate and constrain effective problem-solving. Together, these components provide a comprehensive understanding of human cognition and intellectual functioning.